Pranav Shinde

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EDUCATION

Shivaji University 2020 – 2023
Sangli, Maharashtra, India
Bachelor of Science in Computer Science **CGPA: 8.74/10.0**

National Institute of Technology Karnataka 2024 – Present
Surathkal, Karnataka, India
Master of Science in Physics **CGPA: 8.59/10.0**

PROJECTS

Quantum Kernel SVM for Blood Transfusion Classification **March 2026 – April 2026**
Quantum Machine Learning Project (PH874 - Basics of Quantum Computing) *[view repo]*

- Implemented quantum kernel-based SVM using Qiskit with parameterized 2-qubit feature maps encoding data through R_Y and R_Z rotation gates, motivated by the structural parallel between Hilbert space geometry in quantum mechanics and kernel-based classification in machine learning.
- Designed and compared four interaction kernel functions: xy, $e^{(x-y)/\pi}$, $(x^2+y^2)/\pi$, and $\pi(1-x)(1-y)$, analyzing how feature map choice affects decision boundary geometry and classification performance.
- Computed quantum kernel Gram matrices via statevector inner products and trained precomputed-kernel SVC, achieving consistent $\sim 76.5\%$ test accuracy across all quantum kernels.
- Benchmarked against classical kernels (RBF, linear, polynomial) and developed efficient visualization pipeline for decision boundaries using support-vector-only grid evaluation.

Simulation of Self-Propelled Droplet Dynamics **September 2025 – Present**
Master's Thesis *[view repo]*

- Learned and implemented the Lattice Boltzmann Method (LBM) using the Palabos C++ library, developing initial fluid flow simulations as a foundation for modeling active droplet systems.
- Attempted to reproduce results from a Physical Review E study on Belousov-Zhabotinsky droplets; encountered limitations in implementing multi-field coupling across four lattices, highlighting practical challenges in translating theoretical models into simulations.
- Trained a YOLOv8 object detection model on $\sim$500 manually annotated microscopy images (using CVAT) to detect droplets and extract trajectory data for analysis. [view repo]
- Developed a particle-based simulation framework combining Vicsek-type alignment and chemotactic gradient-following to model emergent collective motion in self-propelled droplet systems. [view repo]
- Bridged data-driven trajectory analysis with theory-driven modeling to refine the final approach for studying collective behavior in active matter.

Monte Carlo Simulation of 2D Ising Model **March 2025 – May 2025**
Computational Physics Project (PH755 - Computational Physics) *[view repo]*

- Implemented Metropolis and Wolff cluster algorithms for 2D spin lattice systems in C++ and Python, comparing single-spin vs. cluster update efficiencies near criticality; first experience using code to study a physical system rather than build a product.
- Reproduced theoretical critical exponents (β, γ, ν) through finite-size scaling analysis, validating second-order phase transition behavior.
- Analyzed critical slowing down and autocorrelation times for both algorithms across temperature ranges spanning the critical point.
- Developed automated data pipeline for ensemble averaging and statistical error analysis.

TECHNICAL SKILLS

Programming: C++, Python
Libraries: NumPy, SciPy, Matplotlib, Pandas, Palabos, Qiskit, PyTorch, Ultralytics (YOLOv8)
Methods: Monte Carlo Sampling, Lattice Boltzmann Method, Particle-Based Simulation, Machine Learning, Statistical Analysis
Tools: Git, Linux, VS Code, Jupyter, Docker, CVAT, LaTeX
Languages: Marathi (Native), English (Professional), Hindi (Professional)